

AES, Mix Column

Date: _____

$$= \begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} \begin{bmatrix} d4 \\ bf \\ 5d \\ 30 \end{bmatrix} = \begin{bmatrix} 02 \times d4 \oplus 03 \times bf \oplus 01 \times 5d \oplus 01 \times 30 \\ 01 \times d4 \oplus 02 \times bf \oplus 03 \times 5d \oplus 01 \times 30 \\ 01 \times d4 \oplus 01 \times bf \oplus 02 \times 5d \oplus 03 \times 30 \\ 03 \times d4 \oplus 01 \times bf \oplus 01 \times 5d \oplus 02 \times 30 \end{bmatrix}$$

2 × d4

Row 1: = $x^1 (11010100)$

$$= x^1 (x^7 + x^6 + x^4 + x^2)$$

$$= x^8 + x^7 + x^5 + x^3$$

$$= x^4 + x^3 + x^1 + x^7 + x^5 + x^3$$

$$= x^3 + x^5 + x^4 + x^1 = 10110010$$

→ 03 × bf

$$= x^1 \cdot 1 (x^7 + x^5 + x^4 + x^3 + x^2 + x^1 + 1)$$

$$= x^8 + x^6 + x^5 + x^4 + x^3 + x^2 + x^1 + x^7 + x^5 + x^4 + x^3 + x^2 + x^1 + 1$$

$$= x^8 + x^7 + x^6 + 1 = x^7 + x^3 + x^1 + x^7 + x^6 + 1$$

$$= x^7 + x^6 + x^4 + x^3 + x^1 + 1 = 11011011$$

→ 01 × 5d (this^{5d} will stay same as multiplied by 1)

$$= 01011101$$

→ 01 × 30 (this³⁰ will also stay same)

$$84 \text{ } 2^8$$

$$= 00110000$$

→ XOR: 10110010

$$\begin{array}{r} \oplus 11011011 \\ \oplus 01101001 \\ \oplus 01011101 \\ \oplus 00110100 \\ \oplus 00110000 \\ \oplus 00000100 \end{array}$$

= 04 → first element is 04.

$$d = 13$$

$$\begin{array}{cccc} A & B & C & D \\ 10 & 11 & 12 & 13 \end{array}$$

$$\begin{array}{r} 8 \ 7 \ 2 \ 1 \ 0 \\ 1 \ 0 \ 1 \ 1 \end{array}$$

Row 2:

→ 01 * d4: (this ^{d4} will stay same)

$$= 11010100$$

→ 02 * bf =

$$= x^1 (x^7 + x^5 + x^4 + x^3 + x^2 + x^1 + 1)$$

$$= x^8 + x^6 + x^5 + x^4 + x^3 + x^2 + x^1$$

$$= \cancel{x^8} + \cancel{x^3} + \cancel{x^1} + x^6 + x^5 + \cancel{x^4} + \cancel{x^2} + x^2 + \cancel{x^1}$$

$$= x^6 + x^5 + x^2 + 1 = 01100101$$

→ 03 * 5d =

$$= x^1 \cdot 1 (\cancel{x^7} + x^6 + x^4 + x^3 + x^2 + 1)$$

$$= x^7 + x^5 + \cancel{x^4} + \cancel{x^3} + x^1 + x^6 + \cancel{x^1} + \cancel{x^3} + x^2 + 1$$

$$= x^7 + x^6 + x^5 + x^2 + x^1 + 1$$

$$= 11100111$$

→ 01 * 30 (this will become 30 when multiplied with 02)

$$= 00100000$$

→ XOR: 11010100

$$\oplus 01100101$$

$$10110001$$

$$\oplus 11100111$$

$$01010110$$

$$00100000$$

$$01100110 = 66$$

Date: _____

→ Row 3:

→ 01 * d4: (will result in d4)

$$= 11010100$$

→ 01 * bf = 10111111

→ 02 * 5d:

$$= x^1 (x^6 + x^4 + x^3 + x^2 + 1)$$

$$= x^7 + x^5 + x^4 + x^3 + x^1 = 1011010$$

→ 03 * 30:

$$= x^1 \cdot 1 (x^5 + x^4) = x^6 + x^5 + x^3 + x^4$$

$$= 01010000$$

→ After XOR = 181

$$\begin{array}{r} 11010100 \\ 10111111 \\ \hline 01101011 \end{array}$$

$$\begin{array}{r} 01101011 \\ \text{XOR } 10111111 \\ \hline 11010001 \end{array}$$

$$\begin{array}{r} 11010001 \\ \text{XOR } 01010000 \\ \hline 10000001 \end{array}$$

$$= 81$$

⇒ Row 4:

→ 03 * d4:

$$= x^1 \cdot 1 (x^7 + x^6 + x^4 + x^2)$$

$$= x^8 + x^7 + x^5 + x^3 + x^1 + x^6 + x^7 + x^2$$

$$= x^8 + x^6 + x^5 + x^3 + x^1 + x^2$$

$$= x^6 + x^5 + x^3 + x^1 + 1$$

$$= 01100111$$

→ 01 * bf: (will result in bf)

$$= 10111111$$

→ 01 * 5d: (will result in 5d)

$$= 01011101$$

→ 02 * 30

$$= x^1 (x^3 + x^4)$$

$$= x^4 + x^5$$

$$= 01100000$$

$$= \begin{bmatrix} 04 \\ 66 \\ 81 \\ e5 \end{bmatrix}$$

$$\begin{array}{r} 01100111 \\ 10111111 \\ \hline 11011000 \end{array}$$

$$\begin{array}{r} 11011000 \\ 01011101 \\ \hline 10000101 \end{array}$$

$$\begin{array}{r} 10000101 \\ 01100000 \\ \hline 11100101 \end{array}$$

$$= e5$$